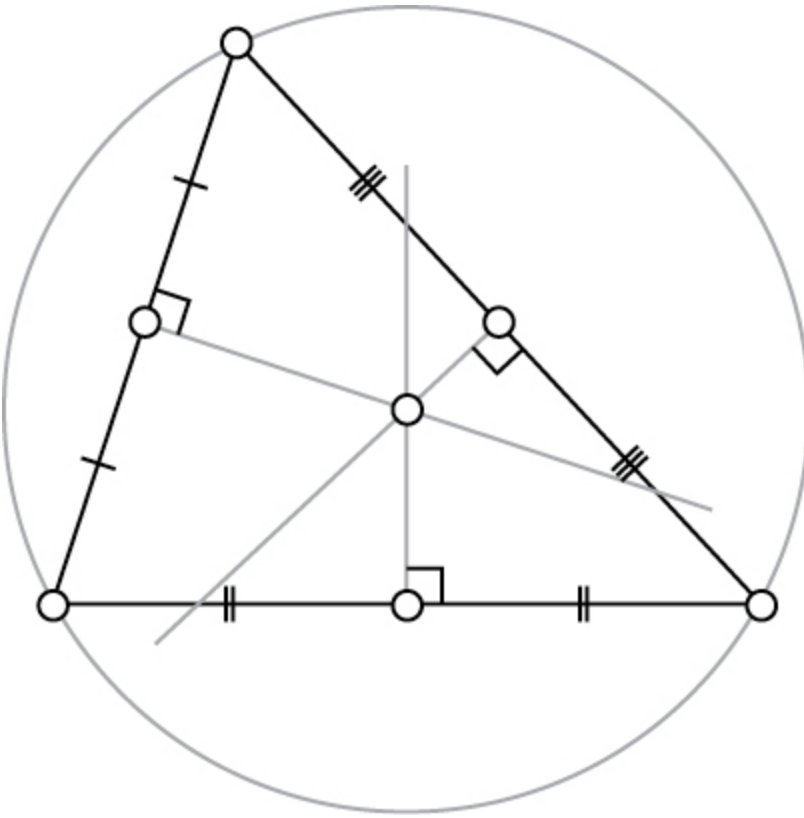




[Figure 7-6:](#) Triangle circumcenter (source: Wikimedia Commons)

Notice that the lines do all meet, again in a point that is not the centroid, but is often inside the triangle. This point has another interesting property: it's the center of the unique circle that goes through all three vertices of our triangle. Here is another of the rich phenomena related to triangles: every triangle has one unique circle that goes through all three of its points. This circle is called the *circumcircle* because it is the circle that circumscribes the triangle. The algorithm we just outlined finds the center of that circumcircle. For this reason, the point where all three of these lines meet is called the *circumcenter*.

Like the centroid, the circumcenter is a point that could be called the center of a triangle, but they are not the only candidates—an encyclopedia at <https://faculty.evansville.edu/ck6/encyclopedia/ETC.html> contains a list of 40,000 (so far) points that could be called triangle centers for one reason or another. As the encyclopedia itself says, the definition of a triangle center is one that “is satisfied by infinitely many objects, of which only finitely many will ever be published.” Remarkably, starting with three simple points